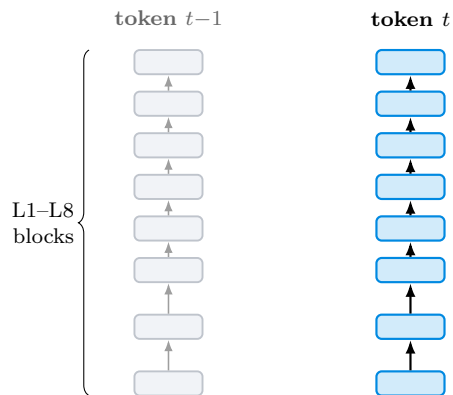


Inference-time information flows

Depth recurrence OFF

Vanilla transformer

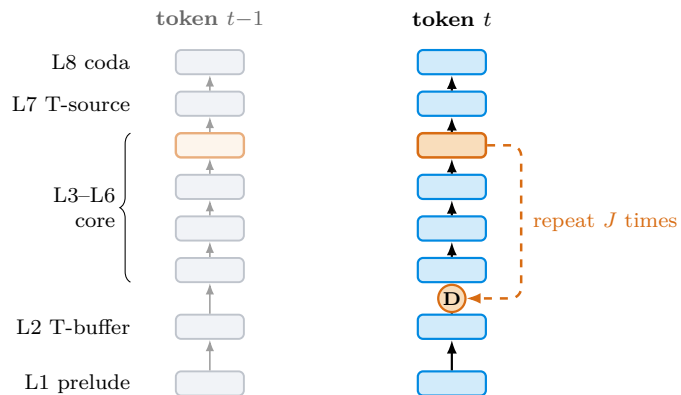
Each token runs L1–L8 once; no state is carried between tokens.



Depth recurrence ON

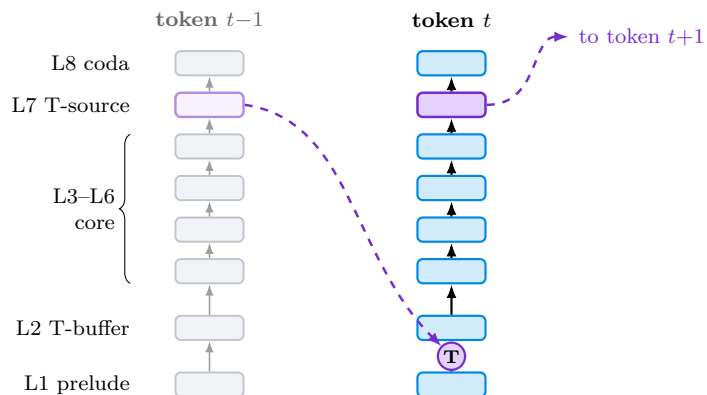
Depth-recurrent / looped LM

The core (L3–L6) runs J times per token; its output re-enters before L3. Depth state is discarded at the next token.



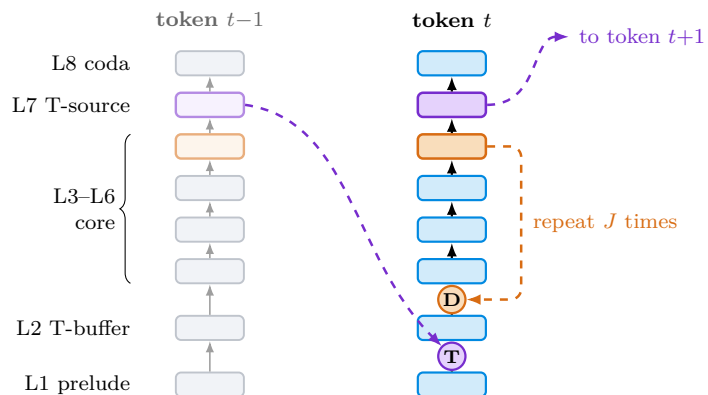
Temporally recurrent

Token $t-1$'s T-source output (L7) is read by token t after L1, so the chain runs through every earlier token.



Depth + temporal hybrid

Temporal state crosses tokens; depth state loops only within the current token.



token $t-1$ layer
→ within-token flow

token t layer
- - -> **D** depth read (loop)

writes depth state
- - -> **T** temporal read

writes temporal state
- - -> state passed to next token